

Interval Numbers: A Formal Algebraic Framework for Indeterminate Forms

Norbert Nopper

Foundation Rule (Rule I):

$$0 \cdot \infty = \Omega := [0, \infty]_{in}$$

Within the *interval-number framework* introduced in this work, the indeterminate form $0 \cdot \infty$ is identified with the **foundation interval** $\Omega := [0, \infty]_{in}$, with companion $-\Omega := [-\infty, 0]_{in}$ for $0 \cdot (-\infty)$ (Rule II). See Section 3.4 for the formal statement, definition, and derivation.

Abstract

The expressions $0 \cdot \infty$ and $0 \cdot (-\infty)$ are classical indeterminate forms that resist standard algebraic manipulation, even under extended real arithmetic. This work introduces *interval numbers*, a formal algebraic framework that represents indeterminate forms as closed intervals in the extended real line $\mathbb{R}$. The interval $[x_0, x_1]_{in}$ is treated as a first-class algebraic object whose endpoints are *assigned by formal rules motivated by directional limit witnesses* rather than derived as a complete classification of all admissible limits. The two foundation rules identify $0 \cdot \infty = \Omega := [0, \infty]_{in}$ and $0 \cdot (-\infty) = -\Omega := [-\infty, 0]_{in}$, naming the resulting interval numbers as a notational shorthand. The rules adopt a *directional* (one-sided) convention that refines the two-sided assignment $[-\infty, \infty]$ of containment-set (cset) arithmetic and is the precise choice under which the sign-reflection identity $\Omega = -1 \cdot (-\Omega)$ carries algebraic content. Operations are defined by interval-arithmetic-style hulls, extended by an explicit value map $\mathcal{V}$ that handles indeterminate endpoint products, and (for exponentiation) by an image-set hull on a stated admissible domain. The main proven result is that $(\mathcal{I}, \cdot)$ is a *unital magma* with identity $[1, 1]_{in}$ —closed under multiplication, but neither associative (an explicit counterexample is given) nor monoidal. Addition, subtraction, multiplication, reciprocal, division, and absolute value are total on $\mathcal{I}$; exponentiation is partial on a precisely stated admissible domain. Worked examples and parametric limit witnesses for each classical indeterminate form (including $0 \cdot \infty$, $\infty - \infty$, $0/0$, ∞/∞ , 0^0 , 1^∞ , ∞^0) illustrate consistency with the chosen rules; no global optimality or tightness theorem is claimed. A C++ reference implementation with an accompanying unit-test suite is provided as a conformance artefact.

Keywords: indeterminate forms, interval arithmetic, extended real numbers, algebraic structure, interval algebra.

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1 Introduction

The treatment of indeterminate forms represents a foundational challenge in mathematical analysis. Within the classical extended real number system $\overline{\mathbb{R}}$, certain expressions—such as $0 \cdot \infty$, $\infty - \infty$, and $\frac{0}{0}$ —cannot be assigned definite values. In standard practice, such expressions are left undefined or handled through case-by-case limit analysis. However, this approach lacks generality: different sequences approaching an indeterminate form may exhibit different limiting behavior, yet no unified algebraic framework exists for treating them.

1.1 Motivating Example

Consider the intuitive observation that if $x = -1 \cdot (-x)$ holds in standard arithmetic, one might expect the following identity to remain valid under appropriate extensions:

$$0 \cdot \infty = -1 \cdot (0 \cdot (-\infty)).$$

A naive algebraic manipulation under the associative law yields:

$$-1 \cdot (0 \cdot (-\infty)) = (-1 \cdot 0) \cdot (-\infty) = 0 \cdot (-\infty),$$

which provides no new information. The fundamental obstacle is that indeterminate forms do not admit standard algebraic treatment.

1.2 Limit-Based Investigation

Standard limit analysis confirms the difficulty. Two sequences both fitting the symbolic form $0 \cdot \infty$ produce different limits:

$$a_n = \frac{1}{n}, \quad b_n = n^2 \Rightarrow a_n b_n = n \rightarrow \infty,$$

$$a_n = \frac{1}{n^2}, \quad b_n = n \Rightarrow a_n b_n = \frac{1}{n} \rightarrow 0.$$

A single real-valued limit cannot capture both behaviours, yet both arise from the same symbolic form. This motivates a representation in which $0 \cdot \infty$ stands for the *range* of attainable limit values rather than a single number.

1.3 Proposed Resolution

This work proposes a resolution by introducing **interval numbers** as formal mathematical objects. Rather than treating $0 \cdot \infty$ as undefined, it is represented as $[0, \infty]_{in}$ —an interval whose endpoints span the limits realizable by directional sequences $a_n \cdot b_n$ with $a_n \rightarrow 0^+$ and $b_n \rightarrow +\infty$ (see Section 3.4 for the precise convention). This foundation interval is denoted $\Omega := [0, \infty]_{in}$, with companion shorthand $-\Omega := [-\infty, 0]_{in}$. This approach makes the basic arithmetic operations (addition, subtraction, multiplication, reciprocal, division, absolute value) total on $\mathcal{I}$, and provides an exponentiation operation defined on a precisely stated admissible domain (Section 4.7).

1.4 Contributions

The main contributions of this work are:

1. A formal definition of **interval numbers** as a mathematical structure for representing indeterminate forms, founded on a *directional* rule $0 \cdot \infty = [0, \infty]_{in}$ that strictly refines the two-sided assignment $[-\infty, \infty]$ of containment-set arithmetic [12, 13] (Section 2).
2. A complete specification of the basic arithmetic operations (total on $\mathcal{I}$) and an exponentiation operation on a stated admissible domain, all following the interval-arithmetic principle of taking hulls of attainable values, extended by an explicit value map for indeterminate endpoint products.
3. A proof that interval numbers form a unital magma under multiplication, with a concrete counterexample demonstrating non-associativity.
4. Demonstration of internal consistency through limit-based justifications of interval bounds.
5. A functional C++ reference implementation accompanied by a unit test suite.

Scope and main results. To distinguish what is *proven* from what is *assigned by convention* or *illustrated by example*, this paper takes the following positions:

- **Definitions and conventions** (Sections 3 and 4): the foundation rules I and II (Section 3.4), the value maps $\mathcal{V}$ and $\mathcal{V}^\wedge$ (Sections 4.1 and 4.7), and the convex-hull convention for the reciprocal of a zero-spanning interval (Section 4.4) are formal choices, not derivations.
- **Proven results:** closure of $(\mathcal{I}, \cdot)$ and existence of the two-sided identity $[1, 1]_{in}$ (Theorem 5.1); failure of associativity (Proposition 5.5).
- **Example-based limit witnesses** (Sections 3.6 and 6.3): explicit and parametric directional sequences exhibiting representative endpoint and interior values for each foundation rule and classical indeterminate form.
- **Implementation conformance:** the C++ reference implementation matches the stated definitions on the cases exercised by the unit tests; it is a conformance artefact, not independent evidence for the mathematical claims.
- **Not claimed:** a global soundness or optimality (tightness) theorem covering all composite expressions; full distributivity; closure of exponentiation outside its admissible domain.

1.5 Organization

The remainder of the document is organized as follows. Section 2 reviews related work in interval arithmetic, extended reals, and indeterminate forms. Section 3 establishes the formal definition and foundation rules. Section 4 defines the algebraic operations. Section 5 discusses the algebraic structure. Section 6 presents worked examples. Section 7 concludes with future research directions. A reference C++ implementation accompanies the paper.

2 Related Work

This section situates the contribution within three relevant research strands: interval arithmetic, extended real numbers, and the classical theory of indeterminate forms.

2.1 Interval Arithmetic

Classical interval arithmetic, formalized by Moore [6] and surveyed in modern references [7, 8] including the IEEE 1788-2015 standard [9], provides methods for computing with sets of real numbers represented as closed intervals. The fundamental motivation—controlling rounding error in numerical computation—differs from the focus of the present work on algebraic closure of indeterminate forms. However, the operations on interval numbers introduced here follow the same underlying principle: for the basic arithmetic operations, the definitions take hulls of attainable endpoint values; for exponentiation, a full image-set hull is required (Section 4.7) because endpoint-only formulas fail when the base interval crosses zero with an even-integer exponent.

Standard interval arithmetic does not address indeterminate forms directly; it avoids division by zero, for example, by requiring intervals to not contain zero. The present work extends this framework to actively address cases in which standard arithmetic fails.

Several extensions of classical interval arithmetic relax this restriction. *Kaucher (modal) interval arithmetic* [10] admits improper intervals $[a, b]$ with $a > b$ and yields a group structure under addition; *generalized / extended interval arithmetic* [11] allows division by zero-containing intervals via unbounded or split results; the algebraic construction of *wheels* [14] adds a totalized reciprocal at the cost of weakening the field axioms, and the related construction of *meadows* [15]

provides a totalized inverse in a field-like setting. The present framework differs from these in two respects: (i) it works on the *extended* real line $\overline{\mathbb{R}}$ rather than $\mathbb{R}$, so $\pm\infty$ are first-class endpoints; and (ii) its primary aim is the algebraic representation of classical indeterminate forms, not the totalization of division per se.

Closest to the present work is *containment-set (cset) arithmetic*, introduced by Walster [12] and developed by Pryce and Corliss [13], which evaluates expressions over the extended reals by assigning to each operation at each point the set of *all* attainable limits. Indeterminate forms thereby receive set values: the cset of $x \cdot y$ at $(0, +\infty)$ is the full extended line $[-\infty, +\infty]$, because two-sided sequences $x_n \rightarrow 0$ realize every extended-real limit. The present framework departs from cset arithmetic in three substantive respects.

- (i) *Directional foundation rule.* Rule I assigns $0 \cdot \infty$ the half-line $[0, \infty]_{in}$, obtained by restricting the vanishing factor to one-sided approach (Section 3.4)—a strictly tighter interval than the cset $[-\infty, +\infty]$, and the choice that validates the motivating identity $\Omega = -1 \cdot (-\Omega)$ (Section 6.1), which collapses to a tautology under the two-sided assignment.
- (ii) *Algebraic characterization.* Cset arithmetic was designed for rigorous numerical enclosure; the algebraic structure it induces on intervals was not analysed. Here the multiplicative structure is characterized exactly: closure and a two-sided identity hold (Theorem 5.1) while associativity fails by explicit counterexample (Proposition 5.5), locating $(\mathcal{I}, \cdot, [1, 1]_{in})$ precisely at unital magma.
- (iii) *Structural results.* Inclusion monotonicity is proved for all basic operations together with its exact obstruction—the single degenerate divisor $[0, 0]_{in}$ (Proposition 5.3, Remark 5.4)—a statement that has no counterpart in the cset literature.

2.2 Extended Real Numbers

The extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, originating in the work of Riemann and formalized by Hausdorff and others [2, 3], adjoins $\pm\infty$ to $\mathbb{R}$. While $\overline{\mathbb{R}}$ provides a topological completion, algebraic operations remain partially undefined: $0 \cdot \infty$, $\infty - \infty$, and similar expressions lack standard interpretations.

Measure-theoretic conventions [2] sometimes assign values to such forms in specific contexts (e.g., $0 \cdot \infty = 0$ in certain measure-theoretic constructions), but these are context-dependent and do not constitute a unified algebraic theory.

2.3 Indeterminate Forms

The classification of indeterminate forms [1] has been studied extensively in calculus and real analysis. The present work does not dispute the classical categorization; rather, it provides a unified algebraic framework for manipulating these forms once they are expressed as interval numbers.

2.4 Positioning of This Work

The novelty of the contribution lies in combining the operational rigor of interval arithmetic with the topological scope of the extended real line, yielding a formalism in which indeterminate forms are first-class algebraic objects. The individual ingredients—extended-real endpoints, set-valued treatment of indeterminate points, hull-based closure—are each present in the works cited above, most completely in cset arithmetic [12, 13]. What is new here is, first, the *directional* foundation rule $0 \cdot \infty = [0, \infty]_{in}$, which refines the two-sided cset assignment $[-\infty, +\infty]$ and is the precise choice under which the sign-reflection identity $\Omega = -1 \cdot (-\Omega)$ carries algebraic content; and second, the resulting *algebraic-structure analysis*—the unital-magma theorem, the

non-associativity counterexample, the embedding of $\overline{\mathbb{R}}$, and the inclusion-monotonicity result with its exact obstruction—none of which is developed in the enclosure-oriented cset literature. To the author’s knowledge, this directional convention and the algebraic characterization built on it have not previously been treated as a self-contained framework.

3 Interval Numbers: Formal Definition

This section first fixes the arithmetic of the extended real line, then introduces interval numbers as formal mathematical objects, and finally states the foundation rules that govern indeterminate forms.

3.1 Preliminaries: Arithmetic in $\overline{\mathbb{R}}$

Let $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$ denote the extended real line, equipped with the order extension $-\infty \leq r \leq +\infty$ for all $r \in \mathbb{R}$.

The standard operations on $\overline{\mathbb{R}}$ adopted in this work are summarized below.

Defined operations. For $r \in \mathbb{R}$ and $s \in \{-\infty, +\infty\}$:

Operation	Result	Conditions
$r + s$	s	any $r \in \mathbb{R}$, $s \in \{-\infty, +\infty\}$
$\infty + \infty$	∞	
$(-\infty) + (-\infty)$	$-\infty$	
$r \cdot \infty$	∞	$r > 0$
$r \cdot \infty$	$-\infty$	$r < 0$
$r \cdot (-\infty)$	$-\infty$	$r > 0$
$r \cdot (-\infty)$	∞	$r < 0$
$\infty \cdot \infty$	∞	
$\infty \cdot (-\infty)$	$-\infty$	
r/∞	0	any $r \in \mathbb{R}$
$r/(-\infty)$	0	any $r \in \mathbb{R}$

Indeterminate forms. The following expressions are *not* assigned a value in $\overline{\mathbb{R}}$ and are the object of study of this work:

$$0 \cdot \infty, \quad 0 \cdot (-\infty), \quad \infty + (-\infty), \quad \infty - \infty, \quad \frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0^0, \quad 1^\infty, \quad \infty^0.$$

In the standard treatment, computations encountering any of these forms are undefined. The interval-number framework provides a uniform algebraic interpretation.

Interval hull. For any nonempty subset $S \subseteq \overline{\mathbb{R}}$, define the *interval hull*

$$\text{Hull}(S) := [\inf S, \sup S].$$

Throughout the paper, infima and suprema are taken in the extended order on $\overline{\mathbb{R}}$, which is a complete lattice, so $\inf S$ and $\sup S$ exist and lie in $\overline{\mathbb{R}}$ for any nonempty S . For finite S —the case arising in all basic arithmetic operations—the infimum and supremum are attained, i.e. $\text{Hull}(S) = [\min S, \max S]$; the general (infinite) case is needed for the directional limit sets of Proposition 3.5 and for the image-set definition of exponentiation (Section 4.7).

Proposition 3.1 (Well-definedness of the hull). For every nonempty $S \subseteq \overline{\mathbb{R}}$, $\text{Hull}(S)$ is an interval number in the sense of Definition 3.2; equivalently, $\inf S, \sup S \in \overline{\mathbb{R}}$ and $\inf S \leq \sup S$.

Proof. $\overline{\mathbb{R}}$ is a complete lattice: every subset of $\overline{\mathbb{R}}$ has an infimum and a supremum in $\overline{\mathbb{R}}$. For nonempty S , pick any $s \in S$; then $\inf S \leq s \leq \sup S$. Hence $\text{Hull}(S) = [\inf S, \sup S]$ satisfies Definition 3.2. If S is in addition finite, the infimum and supremum are attained, so $\text{Hull}(S) = [\min S, \max S]$. $\square$

The hull operator is the common building block of every operation defined in Section 4: each operation is expressed as the hull of a finite candidate set obtained by applying a value map to the endpoint combinations of its operands.

3.2 Definition of an Interval Number

Definition 3.2 (Interval Number). Let $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ denote the extended real line equipped with the standard ordering. An **interval number** is a closed interval in $\overline{\mathbb{R}}$:

$$[x_0, x_1]_{in} := \{x \in \overline{\mathbb{R}} \mid x_0 \leq x \leq x_1\},$$

where $x_0, x_1 \in \overline{\mathbb{R}}$ and $x_0 \leq x_1$.

Notation. Throughout this work, ∞ denotes $+\infty$; negative infinity is written explicitly as $-\infty$. Let $\mathcal{I}$ denote the set of all interval numbers.

3.3 Point Intervals

Definition 3.3 (Point Interval). A point interval is an interval number of the form $[r, r]_{in} = \{r\}$, which is identified with the element $r \in \overline{\mathbb{R}}$.

This identification provides a natural embedding $\overline{\mathbb{R}} \hookrightarrow \mathcal{I}$, $r \mapsto [r, r]_{in}$.

3.4 Foundation Rules for Indeterminate Forms

The interval-number framework assigns to each indeterminate form a closed interval in $\overline{\mathbb{R}}$. These assignments are formal algebraic conventions, motivated and supported by directional limit witnesses: in each case, the assigned interval is the convex hull of products of sequences in which the factor tending to zero is restricted to a specified sign. The two fundamental rules are stated as symbolic identities and then made precise by definitions of the named foundation intervals.

Rule I (statement). The indeterminate form $0 \cdot \infty$ is identified with a single named object Ω :

$$0 \cdot \infty = \Omega.$$

Rule II (statement). The indeterminate form $0 \cdot (-\infty)$ is identified with $-\Omega$:

$$0 \cdot (-\infty) = -\Omega.$$

Definition 3.4 (Foundation intervals). The symbols of Rules I and II are defined as the following interval numbers:

$$\Omega := [0, \infty]_{in}, \quad -\Omega := [-\infty, 0]_{in}.$$

For the joint full-range indeterminate (arising e.g. from $\frac{0}{0}$ and $\infty - \infty$, see Section 6.3), define additionally

$$\tilde{\Omega} := [-\infty, \infty]_{in}.$$

These symbols are *names for interval numbers*, not new atomic elements of $\overline{\mathbb{R}}$. They inherit all arithmetic from the operations defined in Section 4 and add no new mathematical content; the bracketed forms $[0, \infty]_{in}$, $[-\infty, 0]_{in}$, and $[-\infty, \infty]_{in}$ remain primary throughout this work, with the shorthand used where it improves readability. The symbol Ω is local to this work and is not to be confused with the first uncountable ordinal of set theory or with asymptotic lower-bound notation in computer science.

Derivation. The identification $\Omega = [0, \infty]_{in}$ of Rule I is justified by exhibiting directional sequences whose products attain every value of $[0, \infty]_{in}$, with no sequence (under $a_n \geq 0$, $a_n \rightarrow 0$, $b_n \rightarrow +\infty$) leaving that interval; the parametric witness $a_n = c/n$, $b_n = n$ yields any $c \in [0, \infty)$, while $a_n = 1/n$, $b_n = n^2$ realizes the right endpoint $+\infty$ (Section 3.6). Rule II then follows from Rule I by scalar multiplication in the algebra of $\mathcal{I}$ (Definition 4.1): multiplying Ω by -1 gives

$$-1 \cdot \Omega = -1 \cdot [0, \infty]_{in} = [-1 \cdot \infty, -1 \cdot 0]_{in} = [-\infty, 0]_{in} = -\Omega,$$

so $0 \cdot (-\infty) = -1 \cdot (0 \cdot \infty) = -1 \cdot \Omega = -\Omega$. The dual direction $0 \cdot \infty = -1 \cdot (0 \cdot (-\infty))$ is verified step-by-step in Section 6.1.

Directional convention. Within Rules I and II, the symbol 0 is interpreted as a one-sided (non-negative) approach to zero, written 0^+ where emphasis is required. Concretely, Rule I represents the set of limits

$$\left\{ \lim_n a_n b_n : a_n \geq 0, a_n \rightarrow 0, b_n \rightarrow +\infty \right\},$$

and Rule II inherits the analogous one-sided restriction via the derivation above. Without this directional restriction, the products $a_n b_n$ with two-sided $a_n \rightarrow 0$ would attain every value in $[-\infty, \infty]$ —precisely the containment set assigned to $0 \cdot \infty$ by cset arithmetic [12, 13]—and the algebraic identity that motivates the framework would collapse: with $0 \cdot \infty = 0 \cdot (-\infty) = [-\infty, \infty]_{in}$, the identity $\Omega = -1 \cdot (-\Omega)$ degenerates to the tautology $[-\infty, \infty]_{in} = [-\infty, \infty]_{in}$ and distinguishes nothing. The directional refinement is therefore the point of departure from cset arithmetic, and the tighter foundation intervals it yields are what give the algebraic analysis of Section 5 its content.

This convention is a formal choice, not a derivation; the framework should be read as an algebraic structure on $\mathcal{I}$ whose rules are *consistent with* the stated class of directional limits, rather than as a complete classification of all possible limits of all sequences fitting the symbolic form.

3.5 Geometric Visualization

The two foundation intervals together cover the extended real line, intersecting at the origin:

Foundation intervals in $\overline{\mathbb{R}}$

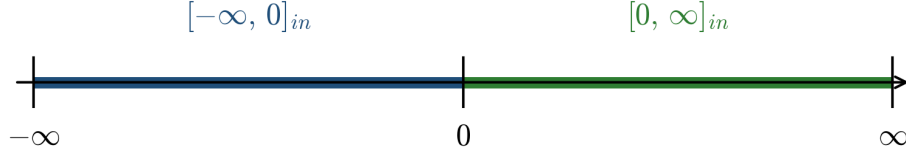


Figure 1: The extended real line $\overline{\mathbb{R}}$ with the two foundation intervals $[-\infty, 0]_{in}$ (blue) and $[0, \infty]_{in}$ (green), corresponding to Rules II and I, respectively. The intervals are not disjoint: they share the point 0.

3.6 Justification for Rules I and II

The interval representations of Rules I and II are justified by exhibiting explicit *directional* sequences (with $a_n \geq 0$) whose products realize a representative spread of limit values within the intervals. The arguments below show that the assigned intervals are *attained*; a separate question—whether they are tight, i.e. whether every interior point is attainable as a limit—is addressed by the parametric witness below.

For $0^+ \cdot \infty$:

- Sequence A: $a_n = \frac{1}{n^2}$, $b_n = n$. Then $a_n b_n = \frac{1}{n} \rightarrow 0$.
- Sequence B: $a_n = \frac{1}{n}$, $b_n = n$. Then $a_n b_n = 1$ for all n .
- Sequence C: $a_n = \frac{1}{n}$, $b_n = n^2$. Then $a_n b_n = n \rightarrow \infty$.

A parametric witness for any $c \in [0, \infty)$ is $a_n = c/n$, $b_n = n$, giving $a_n b_n = c$ for all n ; together with Sequence C, this shows every value of $[0, \infty]_{in}$ is attainable as a limit.

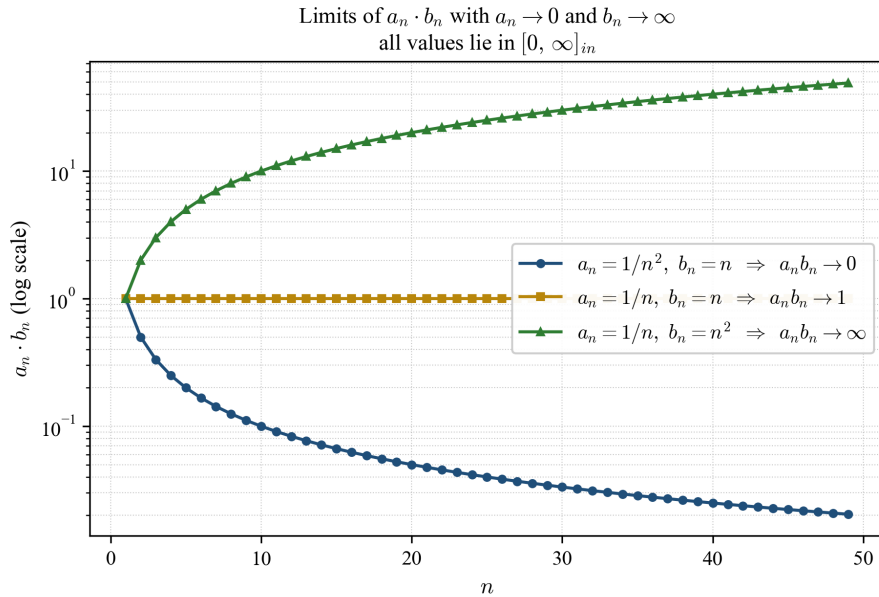


Figure 2: Three sequences $a_n b_n$ with $a_n \rightarrow 0^+$ and $b_n \rightarrow \infty$, converging to 0, 1, and ∞ , respectively (log-scaled vertical axis). All limit values lie within $[0, \infty]_{in}$, consistent with Rule I.

For $0^+ \cdot (-\infty)$:

- Sequence A: $a_n = \frac{1}{n}$, $b_n = -n$. Then $a_n b_n = -1$ for all n .
- Sequence B: $a_n = \frac{1}{n}$, $b_n = -n^2$. Then $a_n b_n = -n \rightarrow -\infty$.

For any $c \in (-\infty, 0]$, the parametric sequences $a_n = -c/n$, $b_n = -n$ (with $a_n \geq 0$ since $c \leq 0$) give $a_n b_n = c$ for all n , completing the witness for $[-\infty, 0]_{in}$.

Proposition 3.5 (Tightness of the foundation intervals). Let

$$L^+ := \{ \lim_n a_n b_n : a_n \geq 0, a_n \rightarrow 0, b_n \rightarrow +\infty, \text{ and the limit exists in } \overline{\mathbb{R}} \},$$

and let L^- be defined analogously with $b_n \rightarrow -\infty$ in place of $b_n \rightarrow +\infty$. Then

$$\text{Hull}(L^+) = [0, \infty]_{in}, \quad \text{Hull}(L^-) = [-\infty, 0]_{in}.$$

Proof. Containment $L^+ \subseteq [0, \infty]$. For each n with n large enough that $b_n > 0$, the product $a_n b_n \geq 0$ since $a_n \geq 0$. Hence every existing limit $\lim_n a_n b_n$ lies in $[0, +\infty]$.

Attainment of $[0, +\infty]$. The witnesses in Section 3.6 show $0, +\infty \in L^+$, and the parametric witness $a_n = c/n$, $b_n = n$ shows $c \in L^+$ for every $c \in [0, +\infty)$. Hence $L^+ \supseteq [0, +\infty]$, and combined with the containment, $L^+ = [0, +\infty]$. Therefore $\text{Hull}(L^+) = [\inf L^+, \sup L^+] = [0, +\infty]$.

The argument for L^- is symmetric: $a_n b_n \leq 0$ for large n since $a_n \geq 0$ and $b_n < 0$; the witnesses of Section 3.6 attain every value of $[-\infty, 0]$. Hence $\text{Hull}(L^-) = [-\infty, 0]$. $\square$

Proposition 3.5 shows that under the directional convention of Section 3.4, the foundation intervals are not arbitrary symbolic choices: they are the unique closed-interval hulls of the directional limit sets, and no strictly smaller closed interval would contain all admissible limit values.

4 Operations on Interval Numbers

All operations follow standard interval arithmetic [6], with explicit treatment of indeterminate forms.

4.1 Multiplication

Definition 4.1 (Interval Multiplication). For interval numbers $I = [x_0, x_1]_{in}$ and $J = [y_0, y_1]_{in}$, define the *candidate set*

$$\mathcal{C}(I, J) := \bigcup_{i,j \in \{0,1\}} \mathcal{V}(x_i, y_j) \subseteq \overline{\mathbb{R}},$$

where the *value map* $\mathcal{V} : \overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \mathcal{P}_{\text{fin}}(\overline{\mathbb{R}}) \setminus \{\emptyset\}$ is defined on each ordered pair of factors by

$$\mathcal{V}(a, b) := \begin{cases} \{a \cdot b\}, & a \cdot b \text{ is defined in } \overline{\mathbb{R}}, \\ \{0, \infty\}, & \{a, b\} = \{0, +\infty\} \text{ (Rule I)}, \\ \{-\infty, 0\}, & \{a, b\} = \{0, -\infty\} \text{ (Rule II)}. \end{cases}$$

The product is then

$$I \cdot J := \text{Hull}(\mathcal{C}(I, J)).$$

Multiplication of finite reals with $\pm\infty$ commutes in $\overline{\mathbb{R}}$, so the value map $\mathcal{V}$ is symmetric in its two arguments. The endpoint sets $\{0, \infty\}$ and $\{-\infty, 0\}$ returned by $\mathcal{V}$ on indeterminate factor pairs are exactly the endpoints of the foundation intervals $\Omega = [0, \infty]_{in}$ and $-\Omega = [-\infty, 0]_{in}$ of Section 3.4; the hull recovers those intervals.

In particular, if no endpoint product $x_i \cdot y_j$ is an indeterminate form, the definition reduces to the classical interval-arithmetic formula

$$I \cdot J = [\min(x_0y_0, x_0y_1, x_1y_0, x_1y_1), \max(x_0y_0, x_0y_1, x_1y_0, x_1y_1)]_{in}.$$

Scalar multiplication. For finite $r \in \mathbb{R}$ identified with the point interval $[r, r]_{in}$, Definition 4.1 specializes to:

$$r \cdot [x_0, x_1]_{in} = \begin{cases} [rx_0, rx_1]_{in}, & r > 0, \\ [rx_1, rx_0]_{in}, & r < 0, \\ [0, 0]_{in}, & r = 0, \ x_0 > -\infty, \ x_1 < +\infty, \\ [-\infty, 0]_{in}, & r = 0, \ x_0 = -\infty, \ x_1 < +\infty, \\ [0, \infty]_{in}, & r = 0, \ x_0 > -\infty, \ x_1 = +\infty, \\ [-\infty, \infty]_{in}, & r = 0, \ x_0 = -\infty, \ x_1 = +\infty. \end{cases}$$

The four $r = 0$ subcases follow from $\mathcal{V}$ applied to the factor pairs $(0, x_0)$ and $(0, x_1)$. For *infinite* scalars $r \in \{-\infty, +\infty\}$ (which may produce indeterminate factor pairs $(r, 0)$ when $0 \in [x_0, x_1]$), no closed-form simplification is asserted; the result is obtained directly from the candidate-set rule of Definition 4.1, e.g.

$$+\infty \cdot [0, 1]_{in} = \text{Hull}(\mathcal{V}(+\infty, 0) \cup \mathcal{V}(+\infty, 1)) = \text{Hull}(\{0, +\infty\}) = [0, +\infty]_{in}.$$

Geometric interpretation. The four endpoint products correspond to the corners of the rectangle $[x_0, x_1] \times [y_0, y_1]$ in the xy -plane:

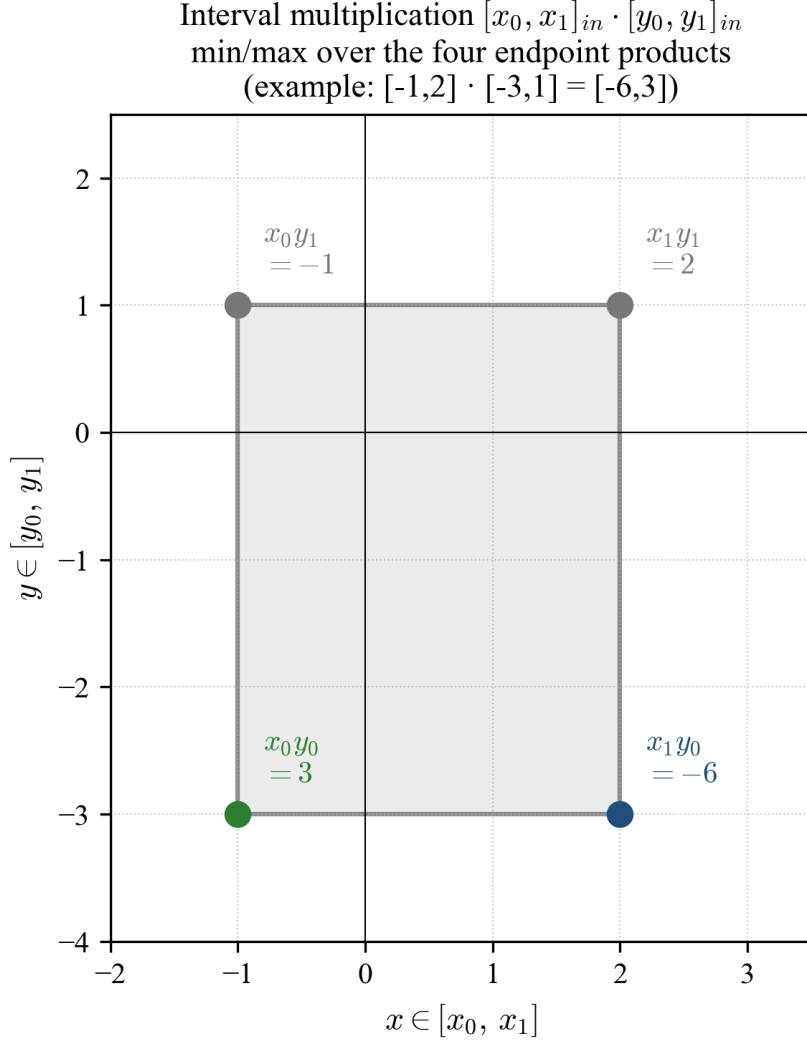


Figure 3: Geometric view of interval multiplication. The product $[x_0, x_1]_{in} \cdot [y_0, y_1]_{in}$ is the closed interval bounded by the minimum and maximum of the four corner products. Example: $[-1, 2]_{in} \cdot [-3, 1]_{in} = [-6, 3]_{in}$.

Examples.

- $(-1) \cdot [0, \infty]_{in} = [-\infty, 0]_{in}$ (validates the consistency between Rule I and Rule II).
- $2 \cdot [1, 3]_{in} = [2, 6]_{in}$.

4.2 Addition

Definition 4.2 (Interval Addition). For interval numbers $I = [x_0, x_1]_{in}$ and $J = [y_0, y_1]_{in}$, define the candidate set

$$\mathcal{C}_+(I, J) := \mathcal{V}_+(x_0, y_0) \cup \mathcal{V}_+(x_1, y_1) \subseteq \overline{\mathbb{R}},$$

where the additive value map $\mathcal{V}_+ : \overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \mathcal{P}_{\text{fin}}(\overline{\mathbb{R}}) \setminus \{\emptyset\}$ is

$$\mathcal{V}_+(a, b) := \begin{cases} \{a + b\}, & a + b \text{ is defined in } \overline{\mathbb{R}}, \\ \{-\infty, \infty\}, & \{a, b\} = \{+\infty, -\infty\}. \end{cases}$$

Then

$$I + J := \text{Hull}(\mathcal{C}_+(I, J)).$$

When neither endpoint sum is indeterminate this reduces to $[x_0 + y_0, x_1 + y_1]_{in}$.

Example. $[0, \infty]_{in} + 5 = [5, \infty]_{in}$. $[0, \infty]_{in} + [-\infty, 0]_{in} = [-\infty, \infty]_{in}$.

4.3 Subtraction

Definition 4.3 (Interval Subtraction). Let the subtractive value map $\mathcal{V}_- : \overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \mathcal{P}_{\text{fin}}(\overline{\mathbb{R}}) \setminus \{\emptyset\}$ be

$$\mathcal{V}_-(a, b) := \begin{cases} \{a - b\}, & a - b \text{ is defined in } \overline{\mathbb{R}}, \\ \{-\infty, \infty\}, & a = b \in \{+\infty, -\infty\}. \end{cases}$$

Define

$$\mathcal{C}_-(I, J) := \mathcal{V}_-(x_0, y_1) \cup \mathcal{V}_-(x_1, y_0), \quad I - J := \text{Hull}(\mathcal{C}_-(I, J)).$$

When no indeterminate form arises this reduces to $[x_0 - y_1, x_1 - y_0]_{in}$.

Example. $[1, 5]_{in} - [2, 3]_{in} = [-2, 3]_{in}$.

4.4 Reciprocal

Definition 4.4 (Reciprocal). Reciprocation is defined on all of $\mathcal{I}$ by the following case analysis on the divisor $J = [y_0, y_1]_{in}$:

$$\frac{1}{J} := \begin{cases} [-\infty, \infty]_{in}, & y_0 = y_1 = 0, \\ [\frac{1}{y_1}, \frac{1}{y_0}]_{in}, & y_0 > 0 \text{ or } y_1 < 0, \\ [\frac{1}{y_1}, +\infty]_{in}, & y_0 = 0, y_1 > 0, \\ [-\infty, \frac{1}{y_0}]_{in}, & y_0 < 0, y_1 = 0, \\ [-\infty, \infty]_{in}, & y_0 < 0 < y_1. \end{cases}$$

The conventions $1/(0^+) = +\infty$ and $1/(0^-) = -\infty$ follow from one-sided real limits; the assignment for $J = [0, 0]_{in}$ is by convention, since neither sign of approach is privileged for a point interval at zero.

When $y_0 < 0 < y_1$, the exact reciprocal is the union of two unbounded branches $[-\infty, 1/y_0]_{in} \cup [1/y_1, +\infty]_{in}$, which is not a single interval. The framework returns its convex hull $[-\infty, \infty]_{in}$, retaining closure within $\mathcal{I}$ at the cost of tightness.

Reciprocal of a zero-spanning interval $J = [y_0, y_1]_{in}$, $y_0 < 0 < y_1$

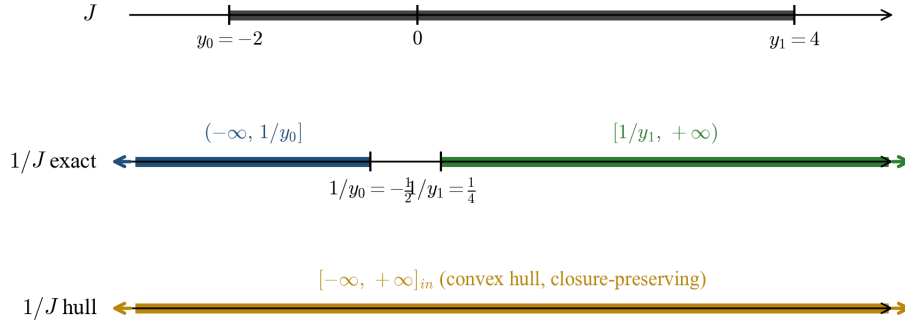


Figure 4: Reciprocal of a zero-spanning interval $J = [y_0, y_1]_{in}$ with $y_0 < 0 < y_1$ (example $J = [-2, 4]_{in}$). The exact reciprocal consists of two unbounded branches $(-\infty, 1/y_0]$ and $[1/y_1, +\infty)$ separated by a gap; the framework replaces this non-convex set with its convex hull $[-\infty, +\infty]_{in}$ to preserve closure within $\mathcal{I}$.

4.5 Division

Definition 4.5 (Interval Division). Division is defined on all of $\mathcal{I} \times \mathcal{I}$ by

$$I \div J := I \cdot \frac{1}{J},$$

evaluated using Definition 4.1 and Definition 4.4. The case analysis of the reciprocal makes division a total operation on $\mathcal{I}$.

Indeterminate form $\frac{0}{0}$. With $I = J = [0, 0]_{in}$, Definition 4.4 gives $1/J = [-\infty, \infty]_{in}$, and then $[0, 0]_{in} \cdot [-\infty, \infty]_{in} = [-\infty, \infty]_{in}$ via Rules I and II in Definition 4.1.

This is consistent with the one-sided limit witnesses

$$\lim_{x \rightarrow 0^+} \frac{x}{x^2} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{x}{x^2} = -\infty,$$

and the parametric witness $a_n = c/n$, $b_n = 1/n$ giving $a_n/b_n = c$ for any $c \in \mathbb{R}$.

Indeterminate form $\frac{\infty}{\infty}$. With $I = J = [\infty, \infty]_{in}$, Definition 4.4 gives $1/J = [0, 0]_{in}$, and then $[\infty, \infty]_{in} \cdot [0, 0]_{in} = [0, \infty]_{in}$ via Rule I in Definition 4.1.

4.6 Absolute Value

Definition 4.6 (Absolute Value). Absolute value is defined by order rather than by endpoint multiplication, so that no indeterminate product is required:

$$|[x_0, x_1]_{in}| := \begin{cases} [x_0, x_1]_{in}, & 0 \leq x_0, \\ [-x_1, -x_0]_{in}, & x_1 \leq 0, \\ [0, \max(-x_0, x_1)]_{in}, & x_0 < 0 < x_1. \end{cases}$$

Negation of $\pm\infty$ is interpreted in $\overline{\mathbb{R}}$ in the standard way.

Examples. $|[-5, 3]_{in}| = [0, 5]_{in}$. $|[-\infty, 0]_{in}| = [0, \infty]_{in}$. $|[0, \infty]_{in}| = [0, \infty]_{in}$.

4.7 Exponentiation

Definition 4.7 (Power). Let $I = [x_0, x_1]_{in}$ and $E = [y_0, y_1]_{in}$. The *admissible domain* of exponentiation is

$$\begin{aligned} \text{dom}(\cdot) := \{ (I, E) \in \mathcal{I} \times \mathcal{I} : & (x_0 > 0) \text{ or } (x_0 \geq 0 \text{ and } y_0 \geq 0) \\ & \text{or } (E = \{n\}, n \in \mathbb{Z}_{\geq 0}) \\ & \text{or } (E = \{n\}, n \in \mathbb{Z}_{<0} \text{ and } 0 \notin I) \}. \end{aligned}$$

On the admissible domain, exponentiation is defined as the closed interval hull of the image set:

$$I^E := \text{Hull}(\mathcal{S}(I, E)),$$

where

$$\mathcal{S}(I, E) := \bigcup_{x \in I, y \in E} \mathcal{V}^\wedge(x, y) \subseteq \overline{\mathbb{R}},$$

and the exponentiation value map $\mathcal{V}^\wedge$ is the *partial* map $\overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \mathcal{P}_{\text{fin}}(\overline{\mathbb{R}}) \setminus \{\emptyset\}$, defined exactly on the pairs falling into one of the two cases below, that extends real-valued exponentiation by handling the indeterminate-form points:

$$\mathcal{V}^\wedge(b, e) := \begin{cases} \{0, \infty\}, & (b, e) \in \{(0, 0), (1, +\infty), (1, -\infty), (+\infty, 0), (-\infty, 0)\}, \\ \{b^e\}, & b^e \text{ is defined in } \overline{\mathbb{R}}. \end{cases}$$

On the admissible domain, every pair $(x, y) \in I \times E$ lies in the domain of $\mathcal{V}^\wedge$, so $\mathcal{S}(I, E)$ is a nonempty subset of $\overline{\mathbb{R}}$ and $I^E = \text{Hull}(\mathcal{S}(I, E))$ is an interval number by Proposition 3.1. Moreover, the hull endpoints are *attained* in $\mathcal{S}(I, E)$: the real-valued continuous map $(x, y) \mapsto x^y$ is monotone in each argument on each of the rectangles obtained by splitting $I \times E$ along $x = 1$ and $y = 0$, so its image is a closed interval, and the finitely many indeterminate-form contributions of $\mathcal{V}^\wedge$ only add isolated endpoint values.

That is, exponentiation is admitted when (i) the base interval is strictly positive, or (ii) the base interval is non-negative and the exponent interval is non-negative (so no 0^y with $y < 0$ arises), or (iii) the exponent is a single non-negative integer (so negative bases are handled by integer powers and zero bases give $0^n = 0$ for $n \geq 1$ and 0^0 via $\mathcal{V}^\wedge$), or (iv) the exponent is a single negative integer and $0 \notin I$ (so 0^n with $n < 0$ does not arise).

The admissible domain is chosen precisely so that every pair $(x, y) \in I \times E$ falls into one of the two cases above; in particular, points 0^y with $y < 0$ are excluded by the domain restriction (i)–(iv). If $(I, E) \notin \text{dom}(\cdot)$ the operation is mathematically undefined; the reference implementation tests membership in the admissible domain explicitly and returns NaN for any pair lying outside it.

Reduction to corner formulas. On regions where $x \mapsto x^y$ and $y \mapsto x^y$ are jointly monotonic, the supremum and infimum of $\mathcal{S}(I, E)$ are attained at corners (x_i, y_j) and Definition 4.7 reduces to a four-corner endpoint formula. This is *not* the case in general; in particular, for a point positive-integer exponent $E = \{n\}$ with $n \in \mathbb{Z}_{>0}$ and a base interval crossing zero, $x \mapsto x^n$ attains its minimum 0 in the interior:

$$[x_0, x_1]_{in}^{\{n\}} = \begin{cases} [x_0^n, x_1^n]_{in}, & 0 \leq x_0 \text{ or } n \text{ odd}, \\ [x_1^n, x_0^n]_{in}, & x_1 \leq 0 \text{ and } n \text{ even}, \\ [0, \max(x_0^n, x_1^n)]_{in}, & x_0 < 0 < x_1 \text{ and } n \text{ even}, \end{cases} \quad (n \in \mathbb{Z}_{>0}).$$

The case $n = 0$ is *not* covered by the formula above: if I contains an indeterminate base for the exponent 0—that is, if $0 \in I$ (giving the pair $(0, 0)$) or if $\pm\infty \in I$ (giving the pair

$(\pm\infty, 0)$)—then $\mathcal{V}^\wedge$ contributes $\{0, \infty\}$ at that point and $I^{\{0\}} = [0, \infty]_{in}$; otherwise every point of I is finite and nonzero, $b^0 = 1$ pointwise, and $I^{\{0\}} = [1, 1]_{in}$. The case $n \in \mathbb{Z}_{<0}$ is admissible only when $0 \notin I$ (case (iv) above); on that region $x \mapsto x^n$ is monotone and the result is

$$[x_0, x_1]_{in}^{\{n\}} = [\min(x_0^n, x_1^n), \max(x_0^n, x_1^n)]_{in}, \quad (n \in \mathbb{Z}_{<0}, 0 \notin I).$$

For strictly positive finite base intervals $I \subseteq (0, +\infty)$, write $x^y = \exp(y \log x)$. The image $y \log x$ over $\log I \times E$ is determined by the standard interval-product hull on these finite quantities, and applying the increasing map $\exp$ yields the corresponding hull for x^y . The only indeterminate closure point that can arise in this case is $1^{\pm\infty}$ (when $1 \in I$ and $\pm\infty \in E$), which is supplemented via $\mathcal{V}^\wedge$; the extended-base indeterminate pair $(\pm\infty, 0)$ does not arise here and is handled by the general image-set definition above.

Indeterminate forms.

$$0^0 = 1^\infty = \infty^0 = [0, \infty]_{in}.$$

Each of these arises as $\mathcal{V}^\wedge$ applied to the corresponding indeterminate-form point. The signed variants admitted by Definition 4.7—namely $1^{-\infty}$ and $(-\infty)^0$ —are also indeterminate and receive the same hull $[0, \infty]_{in}$ by formal assignment via $\mathcal{V}^\wedge$. For $1^{-\infty}$, the parametric sequence $a_n = 1 + (\ln c)/n$, $b_n = -n$ gives $a_n^{b_n} \rightarrow c^{-1}$, attaining every $c^{-1} \in (0, \infty)$ as c varies, so $[0, \infty]_{in}$ is a genuine limit hull. For $(-\infty)^0$, no analogous real-valued limit-witness argument is claimed: non-integer real exponents of negative bases are not real-valued, so $(-\infty)^0 = [0, \infty]_{in}$ is a *formal convention* of the value map $\mathcal{V}^\wedge$ adopted for symmetry with $(+\infty)^0$ and to keep the admissible domain closed under sign reflection of the base.

Justification of $0^0 = [0, \infty]_{in}$:

- Sequence A: $a_n = 1/n$, $b_n = 1/\sqrt{\ln n}$, then $a_n^{b_n} \rightarrow 0$.
- Sequence B: $a_n = 1/n$, $b_n = 1/\ln n$, then $a_n^{b_n} \rightarrow e^{-1}$.
- Sequence C: $a_n = 1/n$, $b_n = -1/\sqrt{\ln n}$, then $a_n^{b_n} \rightarrow \infty$.
- For any $c > 0$, the parametric witness $a_n = 1/n$, $b_n = -\ln c / \ln n$ gives $a_n^{b_n} = c$ for all $n \geq 2$, so every value of $(0, \infty)$ is attained.

5 Algebraic Structure

This section examines the algebraic structure induced by interval-number operations, with emphasis on which classical axioms hold and which fail.

5.1 Closure under Multiplication

Theorem 5.1 (Closure). The set of interval numbers $\mathcal{I}$, equipped with the multiplication operation of Definition 4.1, forms a *unital magma* [5]: it is closed under multiplication, with two-sided identity $[1, 1]_{in}$.

Proof of closure. A magma is a set with a single binary operation that is closed [4, 5]. Let $I = [x_0, x_1]_{in}$ and $J = [y_0, y_1]_{in}$ be interval numbers, with $x_0, x_1, y_0, y_1 \in \overline{\mathbb{R}}$. For each of the four ordered pairs (x_i, y_j) with $i, j \in \{0, 1\}$, the product $x_i \cdot y_j$ is either:

- a defined element of $\overline{\mathbb{R}}$, in which case $\mathcal{V}(x_i, y_j) = \{x_i \cdot y_j\}$ is a singleton; or
- an indeterminate pair $\{x_i, y_j\} = \{0, \pm\infty\}$, in which case Rules I and II assign $\mathcal{V}(x_i, y_j) \in \{\{0, \infty\}, \{-\infty, 0\}\}$, a finite two-element subset of $\overline{\mathbb{R}}$.

In either case $\mathcal{V}(x_i, y_j)$ is a nonempty finite subset of $\overline{\mathbb{R}}$. Hence $\mathcal{C}(I, J) = \bigcup_{i,j} \mathcal{V}(x_i, y_j)$ is a nonempty finite subset of $\overline{\mathbb{R}}$, and by Proposition 3.1, $I \cdot J = \text{Hull}(\mathcal{C}(I, J)) \in \mathcal{I}$. This establishes closure. $\square$

Proof of identity. For $I = [x_0, x_1]_{in}$ with $x_0, x_1 \in \overline{\mathbb{R}}$, the factor pairs in $[1, 1]_{in} \cdot I$ are $(1, x_0)$ and $(1, x_1)$ (each appearing twice). Since one factor is the finite nonzero scalar 1, no indeterminate pair $\{0, \pm\infty\}$ arises, and each product is defined in $\overline{\mathbb{R}}$ with $1 \cdot x_i = x_i$. Hence $\mathcal{C}([1, 1]_{in}, I) = \{x_0, x_1\}$ and

$$[1, 1]_{in} \cdot I = \text{Hull}(\{x_0, x_1\}) = [x_0, x_1]_{in} = I.$$

The argument for $I \cdot [1, 1]_{in}$ is symmetric, since $\mathcal{V}$ is symmetric in its arguments. Uniqueness: if $[e_0, e_1]_{in}$ is also a two-sided identity, then $[e_0, e_1]_{in} = [e_0, e_1]_{in} \cdot [1, 1]_{in} = [1, 1]_{in}$, where the first equality uses that $[1, 1]_{in}$ is a right identity and the second that $[e_0, e_1]_{in}$ is a left identity. Hence $[1, 1]_{in}$ is the unique two-sided identity. $\square$

Proposition 5.2 (Embedding $\overline{\mathbb{R}} \hookrightarrow \mathcal{I}$). The map $\iota : \overline{\mathbb{R}} \rightarrow \mathcal{I}$, $r \mapsto [r, r]_{in}$ is injective and satisfies the following compatibility with $\overline{\mathbb{R}}$ arithmetic on its image. Let $\circ$ denote any of $+$, $-$, $\cdot$ in $\overline{\mathbb{R}}$ and its lift to $\mathcal{I}$ from Section 4. If $r \circ s$ is defined in $\overline{\mathbb{R}}$, then $\iota(r) \circ \iota(s) = \iota(r \circ s)$. If $r \circ s$ is indeterminate, then $\iota(r) \circ \iota(s)$ is the corresponding foundation interval $(\Omega, -\Omega, \text{ or } \tilde{\Omega})$.

Proof. Injectivity is immediate from Definition 3.3. For multiplication: if $r \cdot s$ is defined in $\overline{\mathbb{R}}$, the four factor pairs of $\iota(r) \cdot \iota(s)$ collapse to a single pair (r, s) with $\mathcal{V}(r, s) = \{r \cdot s\}$, so $\iota(r) \cdot \iota(s) = \text{Hull}(\{r \cdot s\}) = [r \cdot s, r \cdot s]_{in} = \iota(r \cdot s)$. If $\{r, s\} = \{0, +\infty\}$ (resp. $\{0, -\infty\}$), then $\mathcal{V}(r, s) = \{0, \infty\}$ (resp. $\{-\infty, 0\}$), and the hull yields Ω (resp. $-\Omega$). The arguments for $\pm$ are analogous, with $\mathcal{V}_+$ (resp. $\mathcal{V}_-$) on $\{+\infty, -\infty\}$ producing the full-range foundation interval $\tilde{\Omega}$. $\square$

Proposition 5.3 (Inclusion monotonicity). For each of the binary operations $\circ \in \{+, -, \cdot\}$ defined in Section 4, and for all $I, I', J, J' \in \mathcal{I}$,

$$I \subseteq I' \text{ and } J \subseteq J' \implies I \circ J \subseteq I' \circ J'.$$

The same holds for the unary operation $|\cdot|$ (Definition 4.6) and for exponentiation I^E on pairs $(I, E), (I', E')$ that both lie in the admissible domain of Definition 4.7. The reciprocal $1/\cdot$ and division $\div$ are *not* inclusion-monotone in general; the single obstruction is the degenerate divisor $[0, 0]_{in}$, and is described in Remark 5.4 below.

Proof. For each operation $\circ \in \{+, -, \cdot\}$, the result equals the hull of the *full* image set $\mathcal{S}_\circ(I, J) := \bigcup_{x \in I, y \in J} \mathcal{V}_\circ(x, y)$, where $\mathcal{V}_\circ$ is the corresponding value map (Definitions 4.1–4.3); the endpoint-only candidate set has the same hull because each value map is monotone in each argument on the sign-constant subrectangles of $I \times J$, so its extreme values are attained at endpoints, while the indeterminate corners contribute exactly the endpoints of the foundation intervals. If $I \subseteq I'$ and $J \subseteq J'$ then $\mathcal{S}_\circ(I, J) \subseteq \mathcal{S}_\circ(I', J')$, and $\text{Hull}(S) \subseteq \text{Hull}(S')$ whenever $S \subseteq S'$, so $I \circ J \subseteq I' \circ J'$. The same image-set argument applies to $|\cdot|$ (whose value at each point is $|x|$, monotone under inclusion of the operand) and to exponentiation, whose image set $\mathcal{S}(I, E)$ is by construction monotone in (I, E) under set inclusion on the admissible domain. $\square$

Remark 5.4 (Failure of monotonicity for the reciprocal). Unlike the operations above, the reciprocal of Definition 4.4 is not defined as the hull of a pointwise image set: the case $J = [0, 0]_{in}$ is the *convention* $1/[0, 0]_{in} := [-\infty, \infty]_{in}$ (Section 5.3, Limitation 3), which assigns the *widest* interval to the *narrowest* divisor. This breaks inclusion monotonicity at exactly that point. For example,

$$[0, 0]_{in} \subseteq [0, 2]_{in}, \quad \text{yet} \quad \frac{1}{[0, 0]_{in}} = [-\infty, \infty]_{in} \not\subseteq \left[\frac{1}{2}, +\infty\right]_{in} = \frac{1}{[0, 2]_{in}},$$

and likewise $[0, 0]_{in} \subseteq [-2, 0]_{in}$ with $1/[-2, 0]_{in} = [-\infty, -\frac{1}{2}]_{in}$. Since division is defined through the reciprocal (Definition 4.5), $\div$ inherits the same failure. On all other case boundaries of

Definition 4.4 the operation is monotone: every case other than $[0, 0]_{in}$ returns the hull of the pointwise reciprocal image, and an exhaustive check over divisor pairs on the endpoint set $\{-\infty, -2, -1, 0, 1, 2, +\infty\}$ exhibits failures only for the smaller divisor $[0, 0]_{in}$. The obstruction is therefore confined to the single degenerate divisor $[0, 0]_{in}$.

5.2 Failure of Associativity

Proposition 5.5 (Non-associativity). Multiplication on $\mathcal{I}$ is not associative.

Counterexample. Let $a = -1$, $b = 0$, $c = -\infty$ (each viewed as a point interval). Then:

$$b \cdot c = 0 \cdot (-\infty) = [-\infty, 0]_{in} \quad (\text{Rule II}),$$

$$a \cdot (b \cdot c) = -1 \cdot [-\infty, 0]_{in} = [0, \infty]_{in},$$

while

$$a \cdot b = -1 \cdot 0 = 0,$$

$$(a \cdot b) \cdot c = 0 \cdot (-\infty) = [-\infty, 0]_{in} \quad (\text{Rule II}).$$

Hence $a \cdot (b \cdot c) = [0, \infty]_{in} \neq [-\infty, 0]_{in} = (a \cdot b) \cdot c$. □

Non-associativity of multiplication on $\mathcal{I}$: $(a \cdot b) \cdot c \neq a \cdot (b \cdot c)$

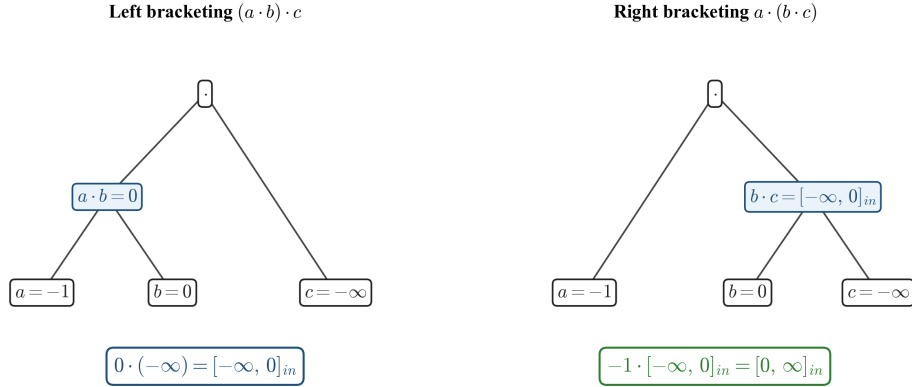


Figure 5: Bracketing trees for the non-associativity counterexample. With $a = -1$, $b = 0$, $c = -\infty$, the left bracketing $(a \cdot b) \cdot c$ resolves the inner product $a \cdot b = 0$ before applying Rule II at the outer node, yielding $[-\infty, 0]_{in}$; the right bracketing $a \cdot (b \cdot c)$ applies Rule II at the inner node and then negates, yielding $[0, \infty]_{in}$. The two results differ.

This counterexample is verified in the implementation by the test `MultiplicationNotAssociative`.

The non-associativity has a precise interpretation: the bracketing of an expression determines *when* an indeterminate form is resolved into an interval; once an interval has been formed, subsequent multiplication propagates uncertainty rather than collapsing it. The failure is therefore not inherited from real multiplication or from classical (Moore) interval multiplication, both of which are associative; it is specific to the rule by which indeterminate symbolic forms are mapped to intervals.

5.3 Limitations of the Structure

Beyond closure and identity, the following stronger algebraic axioms do *not* hold in general:

Axiom	Status	Remark
Closure	holds	Theorem 5.1
Multiplicative identity	holds globally	$[1, 1]_{in}$ is a two-sided identity
Associativity	fails	Proposition 5.5
Multiplicative inverses	fails	intervals containing 0 strictly admit no inverse in $\mathcal{I}$
Distributivity over $+$	fails	sub-distributivity holds in classical interval arithmetic [6]

The structure $(\mathcal{I}, \cdot, [1, 1]_{in})$ is therefore a **unital magma**: closure and a two-sided identity hold, but associativity fails so it is neither a semigroup nor a monoid. Identifying maximal substructures on which stronger axioms recover (e.g., associativity on point-interval subsemigroups, monoidal behaviour on intervals bounded away from zero) is left to future work; see Section 7.

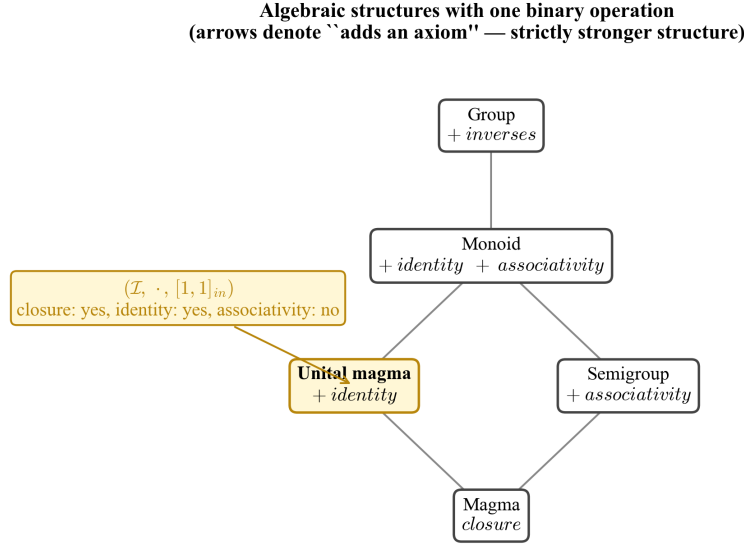


Figure 6: Position of $(\mathcal{I}, \cdot, [1, 1]_{in})$ in the standard hierarchy of algebraic structures with a single binary operation. Each upward edge denotes the addition of an axiom (associativity, identity, or inverses). The structure satisfies closure and identity but not associativity, locating it at **unital magma**; it does not lift to a monoid because of the counterexample of Proposition 5.5.

5.4 Significance

Despite the limitations of Section 5.3, the unital-magma structure is sufficient to support consistent algebraic manipulation of indeterminate forms within bracketed expressions, as demonstrated in Section 6. Crucially, the framework makes the failure of associativity *visible and meaningful* rather than hiding it behind an undefined symbol: bracketing now corresponds to a specific algebraic choice with computable consequences.

6 Worked Examples and Classical Forms

This section illustrates the framework through worked algebraic examples and tabulates the interval-number representations of the classical indeterminate forms together with their limit-based justifications. No application case study from numerical analysis or engineering is claimed

here; the examples document the internal consistency of Definitions 4.1–4.7 with the foundation rules of Section 3.

6.1 Algebraic Consistency

Claim. $0 \cdot \infty = -1 \cdot (0 \cdot (-\infty))$. Equivalently, $\Omega = -1 \cdot (-\Omega)$.

Derivation.

Step	Operation or Rule
$-1 \cdot (0 \cdot (-\infty))$	apply Rule II: $0 \cdot (-\infty) = -\Omega$
$-1 \cdot [-\infty, 0]_{in}$	scalar multiplication (Definition 4.1)
$[-1 \cdot 0, -1 \cdot (-\infty)]_{in}$	operations in $\overline{\mathbb{R}}$
$[0, \infty]_{in} = \Omega$	apply Rule I
$0 \cdot \infty$	$\square$

The interval-number framework therefore validates the intuitive algebraic identity, a result not attainable with naive limit analysis.

6.2 Combined Operations

Example 6.1. $|\frac{0}{0}| + \infty = \infty$.

Derivation.

- $\frac{0}{0} = [-\infty, \infty]_{in}$ (Definition 4.5);
- $|[-\infty, \infty]_{in}| = [0, \infty]_{in}$ (Definition 4.6);
- $[0, \infty]_{in} + \infty = [0 + \infty, \infty + \infty]_{in} = [\infty, \infty]_{in} = \infty$. $\square$

Example 6.2. $\frac{\infty}{\infty} + \infty = \infty$.

Derivation.

- $\frac{\infty}{\infty} = [0, \infty]_{in}$ (see Section 6.3);
- $[0, \infty]_{in} + \infty = [\infty, \infty]_{in} = \infty$. $\square$

6.3 Classical Indeterminate Forms and Their Interval Representations

The following table summarizes the principal indeterminate forms together with their interval-number representations and limit-based justifications.

Form	Interval	Justification
$0 \cdot \infty$	$[0, \infty]_{in}$	Rule I
$0 \cdot (-\infty)$	$[-\infty, 0]_{in}$	Rule II
$\frac{0}{0}$	$[-\infty, \infty]_{in}$	$\lim_{x \rightarrow 0^+} \frac{x}{x^2} = +\infty$ vs. $\lim_{x \rightarrow 0^-} \frac{x}{x^2} = -\infty$
$\infty - \infty$	$[-\infty, \infty]_{in}$	$\lim(n - n^2) = -\infty$ vs. $\lim(n^2 - n) = \infty$
$\frac{\infty}{\infty}$	$[0, \infty]_{in}$	$\lim \frac{n}{n^2} = 0$ vs. $\lim \frac{n^2}{n} = \infty$
0^0	$[0, \infty]_{in}$	parametric $a_n = 1/n$, $b_n = -\frac{\ln c}{\ln n}$ gives $a_n^{b_n} = c$
1^∞	$[0, \infty]_{in}$	parametric $a_n = 1 + \frac{\ln c}{n}$, $b_n = n$ gives $a_n^{b_n} \rightarrow c$
∞^0	$[0, \infty]_{in}$	parametric $a_n = n$, $b_n = \frac{\ln c}{\ln n}$ gives $a_n^{b_n} = c$

These representations are consistent with the definitions in Section 4: for each form, evaluating it via the relevant Section-4 rule reproduces the listed interval. A global soundness theorem covering all composite expressions is not claimed; non-associativity of multiplication (Section 5.2) and convex-hull approximation in the reciprocal (Section 4.4) preclude such a claim in full generality.

Using the shorthand of Section 3.4, the rows of the table can be summarized as $0 \cdot \infty = \Omega$, $0 \cdot (-\infty) = -\Omega$, $0/0 = \tilde{\Omega}$, $\infty - \infty = \tilde{\Omega}$, and $\infty/\infty = 0^0 = 1^\infty = \infty^0 = \Omega$.

6.4 Worked Limits Justifying Selected Intervals

For $\frac{0}{0} = [-\infty, \infty]_{in}$:

$$\lim_{x \rightarrow 0^+} \frac{x}{x^2} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{x}{x^2} = -\infty.$$

For any finite $c \in \mathbb{R}$, the parametric sequences $a_n = c/n$, $b_n = 1/n$ satisfy $a_n/b_n = c$ for all n , demonstrating that every value of $\mathbb{R}$ is attainable.

For $1^\infty = [0, \infty]_{in}$, the endpoints are realized by

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{\sqrt{n}}\right)^{n^2} = 0, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\sqrt{n}}\right)^{n^2} = \infty,$$

and every interior value $c \in (0, \infty)$ is attained by the parametric witness

$$\lim_{n \rightarrow \infty} \left(1 + \frac{\ln c}{n}\right)^n = e^{\ln c} = c.$$

For $\infty^0 = [0, \infty]_{in}$, the endpoints are realized by

$$\lim_{n \rightarrow \infty} n^{-1/\ln \ln n} = 0, \quad \lim_{n \rightarrow \infty} n^{1/\ln \ln n} = \infty,$$

and for every $c > 0$ the parametric witness $a_n = n$, $b_n = (\ln c)/(\ln n)$ gives $a_n^{b_n} = e^{\ln c} = c$ for all $n \geq 2$.

For $0^0 = [0, \infty]_{in}$, for every $c > 0$ the parametric witness $a_n = 1/n$, $b_n = -(\ln c)/(\ln n)$ satisfies $a_n^{b_n} = n^{\ln c / \ln n} = c$ for all $n \geq 2$, while $b_n = \pm 1/\sqrt{\ln n}$ yields the endpoints 0 and ∞ .

7 Conclusion and Future Work

7.1 Summary

This work has introduced **interval numbers** as a formal algebraic framework for representing and manipulating indeterminate forms. By embedding indeterminate expressions into closed intervals of the extended real line $\mathbb{R}$, the basic arithmetic operations (addition, subtraction, multiplication, reciprocal, division, absolute value) are made total on $\mathcal{I}$, and exponentiation is defined on a precisely stated admissible domain. The induced multiplicative structure is a **unital magma**—closed, with two-sided identity $[1, 1]_{in}$, but neither associative nor monoidal—and supports consistent algebraic manipulation while preserving the intuitive interpretation of indeterminate forms as ranges of admissible limit values. A C++ reference implementation accompanied by a unit test suite exercises the framework computationally on representative cases.

7.2 Limitations and Open Questions

To support an honest appraisal, the principal limitations of the framework are stated explicitly:

1. **Multiplication is not associative.** Proposition 5.5 exhibits a concrete counterexample. Consequently, the result of an interval-number expression depends on its bracketing, and the framework only enjoys unital-magma structure—not semigroup, monoid, or group structure—on $\mathcal{I}$ as a whole.
2. **Loss of tightness for division by zero-spanning intervals.** When the divisor strictly contains zero, the natural reciprocal is a union of two intervals; to retain closure within $\mathcal{I}$, the framework returns $[-\infty, \infty]_{in}$, which is sound but not tight (Section 4.4).
3. **Convention behind $\frac{0}{0}$.** The reciprocal of the point interval $[0, 0]_{in}$ is assigned $[-\infty, \infty]_{in}$ by the convex-hull convention of Definition 4.4; the value of $\frac{0}{0}$ then follows internally from Definition 4.5. The choice of $[-\infty, \infty]_{in}$ for the reciprocal of $[0, 0]_{in}$ is a convention motivated by directional limits, not an internal derivation.
4. **No proven distributivity.** Sub-distributivity of multiplication over addition holds in classical interval arithmetic [6]; full distributivity does not transfer, and a precise statement for $\mathcal{I}$ is not given here.
5. **Choice of limit witnesses.** The interval assigned to each indeterminate form is justified by exhibiting limit sequences whose values reach (or approach) the endpoints. A proof that the chosen interval is *optimal*—i.e., that no admissible value lies strictly outside it—is informal and rests on standard real-analytic intuition rather than a fully formal argument.
6. **Floating-point realization.** Any numerical realization (such as the reference implementation) inherits the rounding behaviour of its arithmetic substrate. The mathematical claims are stated in $\overline{\mathbb{R}}$; floating-point realizations are approximate.

These limitations delineate the scope of the present contribution and motivate the directions outlined in Section 7.3.

7.3 Future Directions

The interval-number framework opens several avenues for further investigation:

1. **Richer algebraic structure.** Investigate whether interval numbers exhibit additional structure (associativity, distributivity, identities, inverses) under suitable restrictions—for example, on intervals not containing zero, or on the set of point intervals.
2. **Measure-theoretic integration.** Reconcile interval-number operations with measure-theoretic conventions for handling indeterminate forms [2], where context-specific conventions such as $0 \cdot \infty = 0$ are common.
3. **Applications.** Explore applications in:
 - numerical analysis (controlled treatment of overflow and underflow),
 - complex analysis (extension to complex-valued interval numbers),
 - probability and statistics (representing imprecise probabilities).
4. **Extended operations.** Define and characterize additional operations on interval numbers, including:
 - logarithm and exponential,
 - trigonometric and hyperbolic functions,
 - transcendental functions and their inverses.

5. **Categorical perspective.** Examine interval numbers from a categorical or topos-theoretic viewpoint to uncover deeper structural properties and connections with related frameworks (fuzzy sets, rough sets, interval-valued probabilities).
6. **Formal verification.** Mechanize the theory in a proof assistant (Coq, Lean, Isabelle) to obtain machine-checked guarantees of soundness for all operations and identities.
7. **Estimation of additional indeterminate intervals.** Other indeterminate forms—possibly arising in specific application domains—could be assigned interval representations through the same limit-based justification methodology employed here.
8. **Closure of exponentiation.** Exponentiation is defined here only on the admissible domain stated in Section 4.7; outside that domain it remains partial. Extending the operation to a total map on $\mathcal{I}$ —either by enlarging the admissible domain through additional value-map cases or by adopting a closure convention analogous to the convex-hull convention used for the reciprocal of a zero-spanning interval—is left for future work.

7.4 Closing Remark

While the immediate significance of the framework remains theoretical, it offers a mathematically principled approach to a long-standing problem in real analysis. The unification of interval arithmetic and extended-real semantics provides a foundation upon which further algebraic, computational, and applied investigations can be built.

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